SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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॥वसुधैव कुटुम्बकम्॥

CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 23

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Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.

Learning Objectives : To navigate and retrieve records using ResultSet methods.

Theory : A ResultSet object contains data retrieved from a database after executing a SELECT query. It acts like a cursor that points to one row at a time. The next() method moves to the next row, while methods such as first(), last(), previous(), and absolute() can be used with scrollable ResultSets. Methods such as getString(), getInt(), and getDouble() retrieve column values.

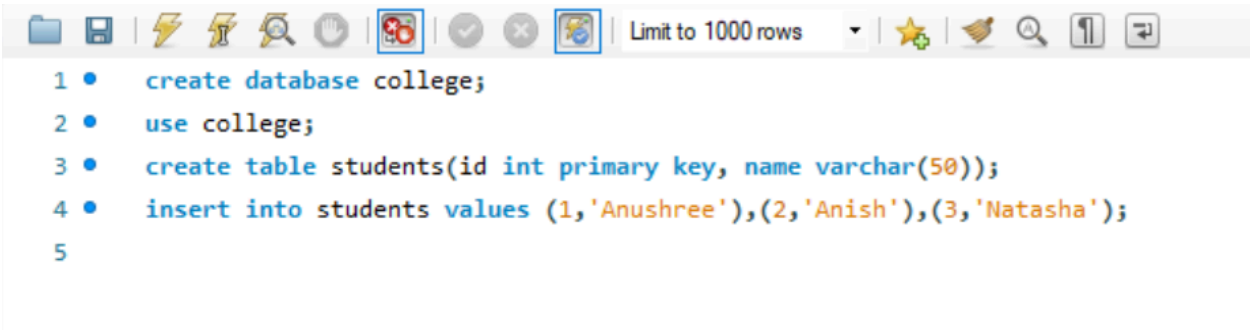
Exercise :

1. Display database records one by one using ResultSet.

```
import java.sql.*;

public class ResultSetDemo {
    public static void main(String[] args) throws Exception {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/college", "root", "anushree");
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("select * from students");
        while (rs.next()) {
            System.out.println(rs.getInt("id") + " " +
                rs.getString("name"));
        }
        rs.close(); st.close(); con.close();
    }
}
```

In SQL :



OUTPUT :



2. Retrieve employee ID, name, department, and salary sequentially.

```
import java.sql.*;

public class EmployeeRecords {
    public static void main(String[] args) throws Exception {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/college", "root", "anushree");
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("select * from employee");
        while (rs.next()) {
            System.out.println("ID: " + rs.getInt("emp_id"));
            System.out.println("Name: " + rs.getString("name"));
            System.out.println("Department: " +
rs.getString("department"));
            System.out.println("Salary: " + rs.getDouble("salary"));
            System.out.println("--------------------");
        }
        rs.close(); st.close(); con.close();
    }
}
```

In SQL :

```
5  create table employee(emp_id int primary key, name varchar(50),
6  department varchar(40), salary double);
7  insert into employee values
8  (101,'Amit','IT',45000),(102,'Neha','HR',42000);
```

OUTPUT :

```
PS C:\Users\softlink\Desktop\java code>
-x64\jre\25.0.4.1-win32-x86_64\bin\java.e
dg57eiz.argfile' 'EmployeeRecords'
ID: 102
Name: Neha
Department: HR
Salary: 42000.0
--------------------
```

Outcome :

- Understood the use of ResultSet in JDBC for retrieving database records.
- Learned to navigate through records using methods such as next(), first(), last(), previous(), and absolute().
- Understood how to retrieve column values using getString(), getInt(), and getDouble().
- Learned to display student and employee records sequentially from a database.
- Successfully implemented Java programs to retrieve and display database records using ResultSet.

Conclusion :

This assignment demonstrated the use of ResultSet for navigating and retrieving records from a database using JDBC. Student and employee records were retrieved and displayed sequentially using appropriate ResultSet methods. The assignment helped in understanding cursor movement, column retrieval, and database record processing in Java applications.